

Practical PCB Design

ECEN 3730

Board 1

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Project Overview:

For the timer circuit on this board to be considered working, I used a similar definition as was used in Lab 2. The 555 timer should be powered with 5V V_{CC} and operate in astable mode with an output a square wave with a frequency of roughly 500Hz (between 300 and 1kHz) and a duty cycle of roughly 50% (between 40% and 75%). This output should be able to drive enough current to power 4 red LEDs in series with 4 different resistors between 50 Ω and 10k Ω .

Circuit Plan and Theory:

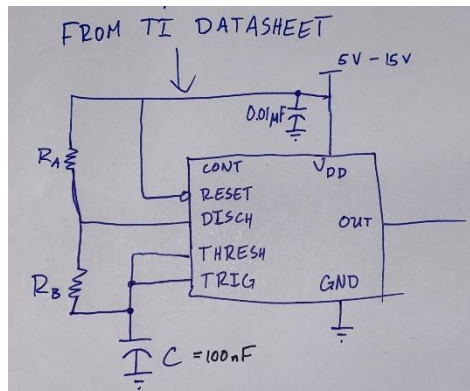


Fig. 1 (Sketch of 555 timer circuit)

Following the same circuit schematic from Lab 2 (seen in Figure 1), inserting a 1 μ F capacitor for C and 1k Ω resistors in for both R_A and R_B , the frequency and duty cycle calculations are as follows:

$$f = \frac{1.44}{(R_A + 2R_B)C} = \frac{1.44}{(1k + 2k)1\mu} = 480Hz$$

$$Duty\ Cycle = f * (0.693(R_A + R_B)C) = 480(.693(2k))1\mu = 66.5\%$$

These specific resistor and capacitor values were chosen based solely on the availability of components for the board. This skewed my expected frequency and duty cycle values away from the desired 500Hz, 50% duty cycle, but is still within the acceptable range as defined in Lab 2.

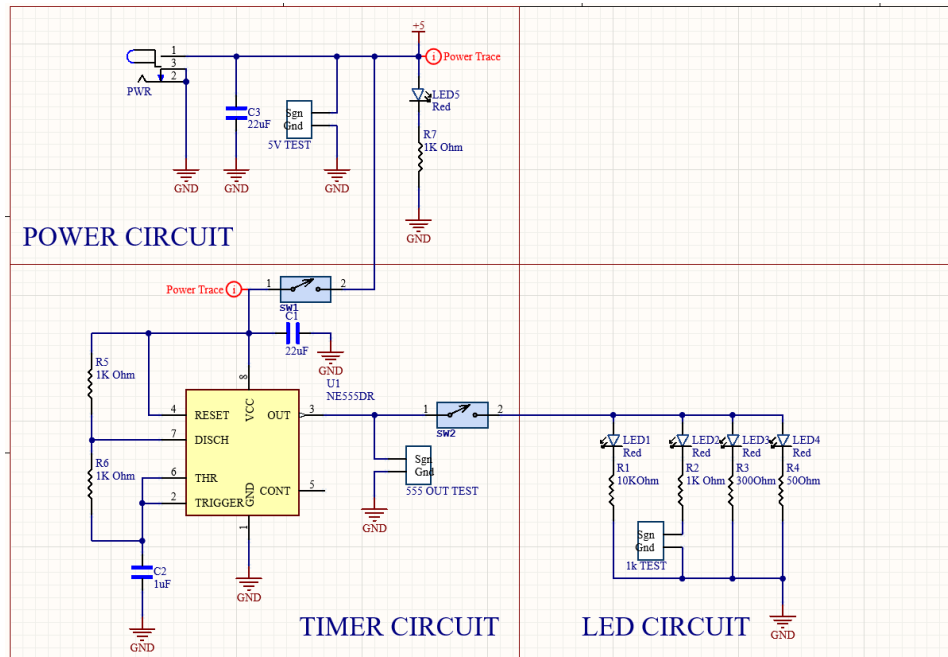


Fig. 2 (Altium schematic of 555 timer circuit)

Figure 2 above shows the full schematic for Board 1 that I built in Altium. The circuit includes three subcircuits: the 555 timer (based on the hand-drawn sketch in Figure 1), the power supply circuit, and the four red LEDs with different series resistances. The power circuit has a decoupling capacitor (C3), a power indicator (LED5 in series with a 1k Ω resistor), and a test point in parallel with the switch to the 555 timer circuit. The switch was placed to keep the timer and LED circuits isolated from power in the event that there is an issue with the power circuit. LED5 acts as an indicator as to whether the board is receiving power from the barrel jack, and the test point serves as a location to verify the exact voltage the circuit is receiving.

The timer circuit was wired the same as in Figure 1 with an isolation switch and a test point added at the output of the timer IC. I decided to use the slow 555 timer for this board to ensure it could push the current needed to drive all four of the LEDs. The isolation switch keeps power away from the LED circuit and allows me to get an accurate measurement of the IC's output voltage in the zero-load case. This is to ensure the timer is outputting the expected frequency and duty cycle before allowing it to power the LEDs. The 555 timer test point also acts as a measurement point to get an estimate of the current draw of the 1k Ω resistor when the LEDs are powered on. The LED circuit is pretty straightforward; there are four LEDs each connected to a resistance of 50 Ω , 300 Ω , 1k Ω , or

10kΩ. The test point for the 1kΩ resistor was wired incorrectly in the schematic, and I will discuss the effect of this later on.

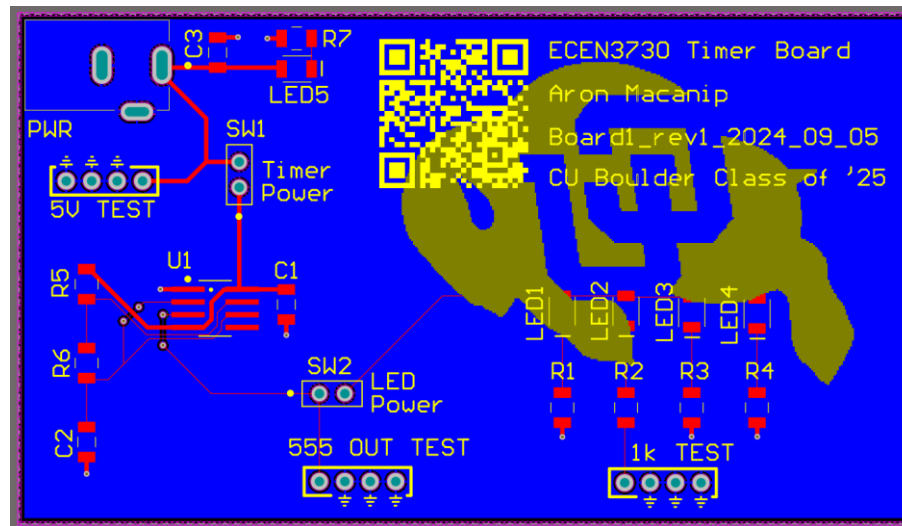


Fig. 3 (Altium PCB layout for Board 1)

Figure 3 shows the final PCB layout of Board 1. 20mil traces were used for any trace which connects components to 5V power or to the ground plane. 6mil traces were used for traces between signal nets. C1 and C3 were both kept close to the 555 timer IC and the power jack respectively in order to keep the trace inductance as small as possible. All ground plane vias were kept close to their respective ground pads for the same reason, giving each component the shortest reasonable return paths. All non-grounding vias were placed in such a way that the signal path across the bottom layer of the board was as short as practical to limit the amount of signal noise to/from the return plane.

I changed the labels for each of the test points and switches on the silk screen layer to make it clear what function each one served. For future boards, I would probably also add a silk screen label for the power indicator LED, showing what voltage the board is supposed to be receiving/putting out to the rest of the circuit, and another near the IC to indicate what IC should be soldered in. On a more complex board with multiple ICs, I think having a clearer label more than simply “U1” could be helpful as long as space on the board allows for it.

Physical Board:

Figure 4a shows the bare physical board printed from the PCB layout seen in Figure 3. Figure 4b shows the board fully soldered. Initially, I had connected each of the three test points on this board in series with the circuits they were meant to be testing rather than in parallel with the node to measure the voltage at that node. This was caught during my TA check off, and I fixed two of the three test points, but the test point for the $1k\Omega$ resistor managed to slip past me and the TA. This can be seen a little more obviously in Figure 4c. Fortunately, this was only a soft error and an easy fix. I just soldered some extra header pins into the holes meant for testing and used a flag to connect the resistor to ground. The black header pins serve to indicate where to connect the flag to ground the $1k\Omega$ resistor, and the yellow header pins serve as a connection point for the scope probe. This error made testing the voltage across the resistor slightly more difficult, as the probe had to be held in place manually, but the overall functionality of the circuit was unaffected.

After all the LEDs were connected, I was able to compare the brightness with the different LED values. For use as indicators, the $10k\Omega$ and 50Ω resistors each made the LED too dim and too bright respectively. The $1k\Omega$ and 300Ω resistors provided their respective LEDs with a reasonable amount of current to act as indicator lights.

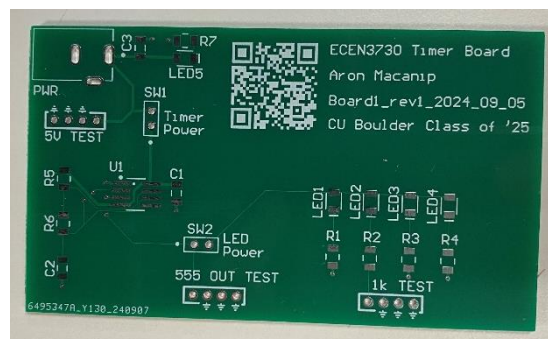


Fig. 4a (Board 1 with no soldered components)

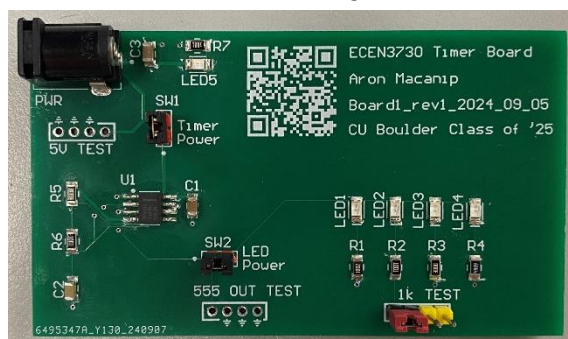


Fig. 4b (Board 1 fully soldered)

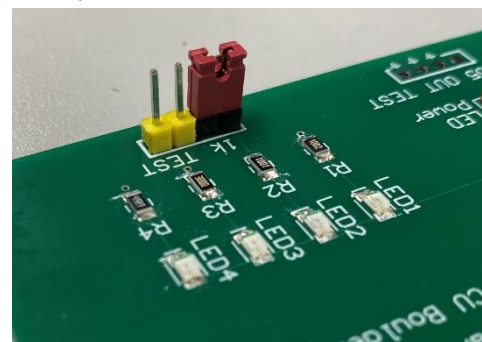


Fig. 4c (Close-up of $1k\Omega$ test point)

Figures 5a through 5d show different measurements of the 555 timer's output with the four LEDs disconnected. As seen in Figure 5a, the measured output frequency and duty cycle of the 555 timer were approximately 505Hz and 65.8% respectively. These values were slightly off from the 480Hz and 66.5% duty cycle I calculated, but within a very reasonable margin of error. I suspect these discrepancies were most likely due to slight variations in the actual values of each component. Figure 5b shows my measurement of the HIGH voltage output of the 555 timer. This measurement was taken to verify that the timer was outputting the expected voltage of roughly 5V to drive the LED circuit before connecting the LEDs in case there were any issues with the timer circuit.

Figures 5c and 5d show the rise and fall times of the timer's output respectively. The 88 nanosecond rise time was roughly what was expected for the slow 555 timer IC. I may have zoomed in a bit too much on the time axis for the "measure" function of the scope to give the most accurate measurement.

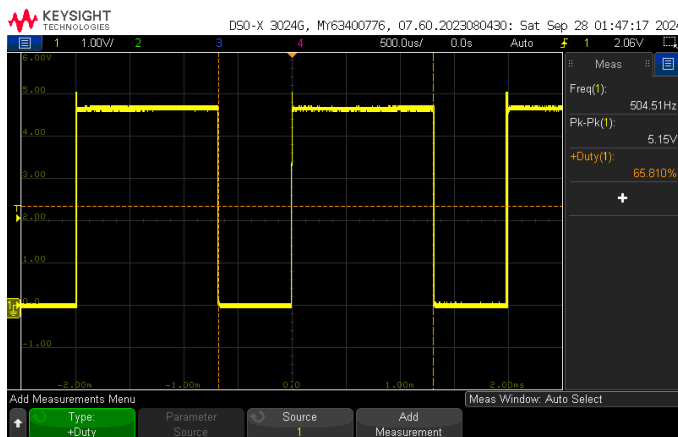


Fig. 5a (Timer frequency and duty cycle measurements)

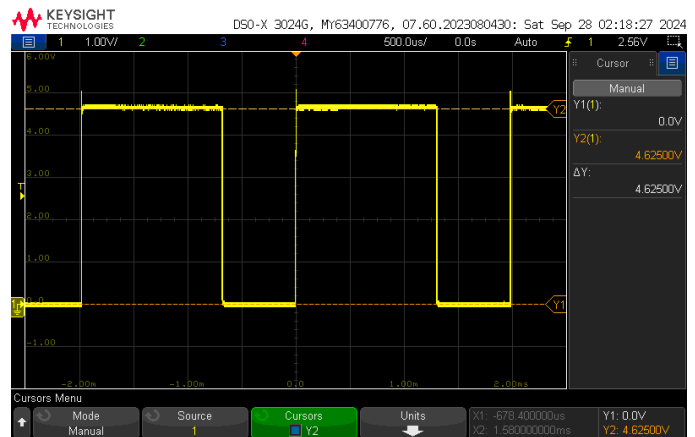


Fig. 5b (Timer HIGH voltage measurement)

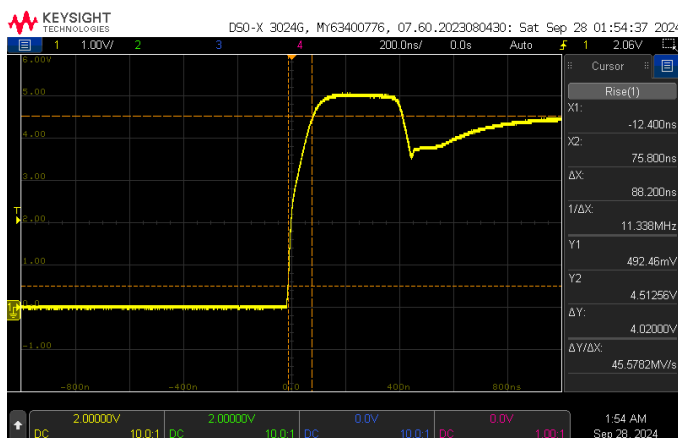


Fig. 5c (Timer rise time measurement)

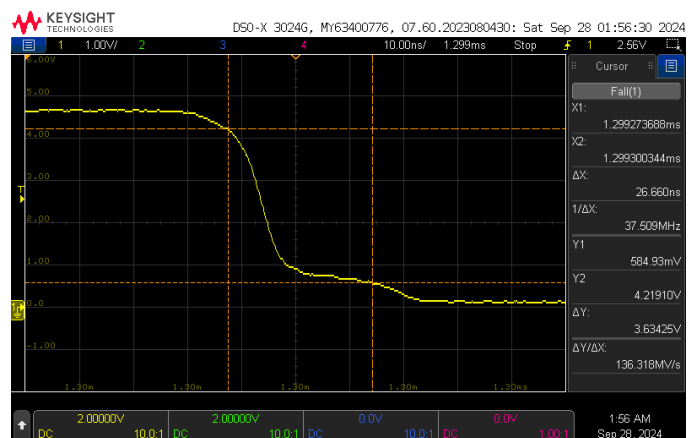


Fig. 5d (Timer fall time measurement)

Figures 6a and 6b show HIGH voltage values for the timer output and the 1kΩ resistor. Using these voltage measurements, we can estimate the amount of current LED2 is drawing as well as how much current the 555 timer is supplying to all four of the LEDs. These calculations are as follows:

$$I_2 = \frac{v_2}{R_2} = \frac{1.88V}{1k\Omega} = 1.88mA$$

$$I_{out} = \frac{(V_{out} - V_{LED})}{R_1 || R_2 || R_3 || R_4} = (3.73V - 2V) \left(\frac{1}{10k\Omega} + \frac{1}{1k\Omega} + \frac{1}{300\Omega} + \frac{1}{50\Omega} \right) \approx 42.3mA$$

Assuming the same approximate voltage drop across each LED of 2V and treating the resistors as a parallel combination, the total current pushed by the 555 timer is estimated to be roughly 42.3mA. The power consumed by the LED circuit is then approximately 158mW. This power consumption is where the difference in the 555 timer's output voltage for the zero-load case and loaded case comes from.

Looking at Figures 6c and 6d, we can see that the introduction of the LEDs also has an effect on the rise and fall times of the timer's output. The rise time with the LED circuit turned on is about 30 nanoseconds shorter than the zero-load case. This is because the timer's HIGH output voltage is noticeably lower when the LED circuit is connected, so it takes less time for the timer to reach its peak voltage. In contrast, the fall time is about 30 nanoseconds longer than the zero-load case. I suspect that this is due to parasitic inductances introduced by the longer traces that connect the LED circuit to the timer's output. The inductance of the LED circuit would resist the quick change in current, lengthening the fall time.

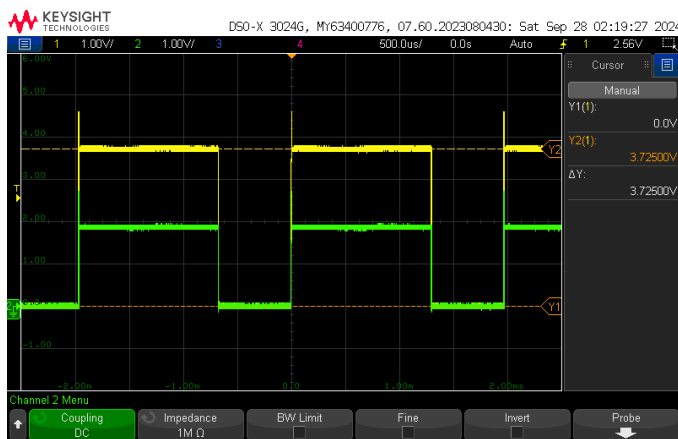


Fig. 6a (Timer HIGH voltage measurement w/ LEDs on)

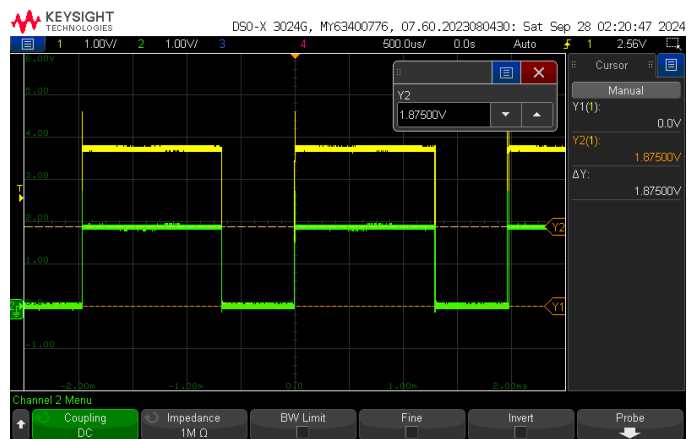


Fig. 6b (R2 HIGH voltage measurement)

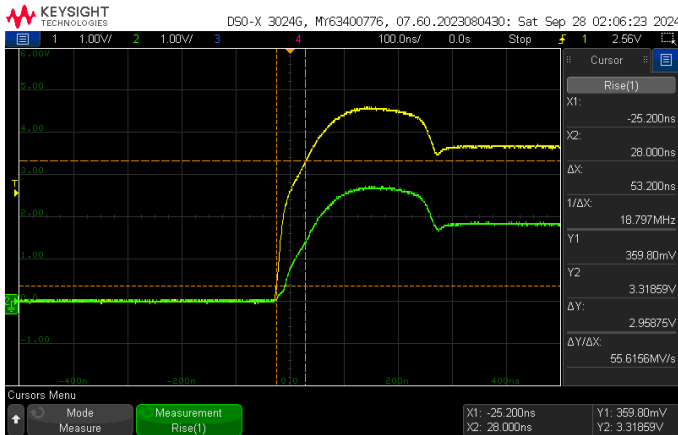


Fig. 6c (Timer rise time measurement w/ LEDs on)

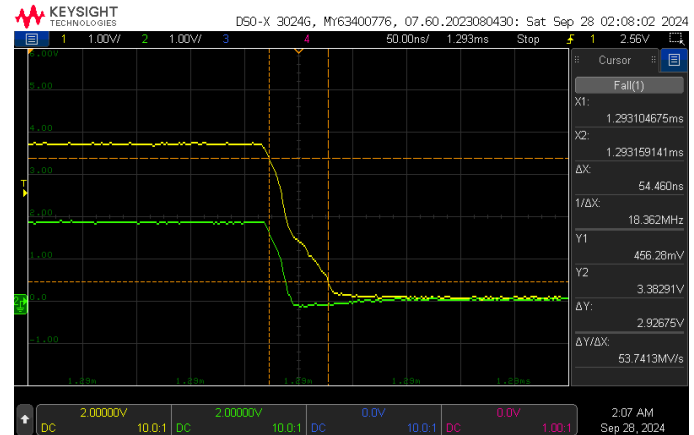


Fig. 6d (Timer fall time measurement w/ LEDs on)

Conclusions:

My biggest take away and learning point with this first board was in the connection of the test points. As mentioned above, my inattention to the 1kΩ resistor test point made resulted in a soft error in my board. In a grander sense, this error served as a great reminder to be diligent and careful every step of the PCB design process.

Soldering of the board went very smoothly, and I am quite satisfied with the way it turned out. I soldered the components in an order based roughly on their 3D profiles, so I wasn't able to test each subcircuit as I soldered. For the next board, I want to try soldering each subcircuit and testing along the way to make sure each circuit has the expected functionality. This was my first time soldering an SOIC component on a board that wasn't just for practice, so I was a little nervous that I would mess up somehow. I did have a little trouble getting the chip tacked on at first, but I quickly managed to get the component in place. I also initially soldered the power indicator LED in the reverse direction. Fortunately, I noticed my mistake before I finished soldering and powered on the board, so I was able to resolder the LED without issue.

On the more logistical side of this project, I found myself rushing to meet certain deadlines, including the design review and board file submission. This was due, in part, to needing to take time to recover from a head injury, but I also feel I could have spent more time before the injury to get ahead. I underestimated how long building out the schematic and board layout in Altium would take me, and also spent time on minor details as a matter of perfectionism, so I still have plenty of room to improve my time-management skills for future boards.